RESISTORS, KIRCHHOFF AND SQUARED RECTANGLES ON ONE PAGE

Every square is a one-ohm resistor

Parallel resistance is a harmonic mean, and Kirchhoff's laws for a network of equal resistors tile a rectangle with squares.

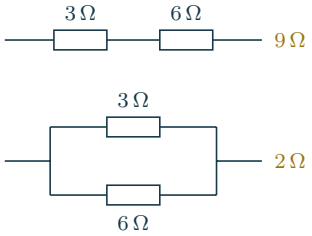
THE RULE

Series adds resistances, parallel adds their reciprocals

In series the same current passes through each resistor and the voltages add; in parallel each resistor feels the same voltage and the currents add, so it is the conductances $1/R$ that add. For n resistors the two totals are means in disguise:

$$R_{\text{series}} = \sum R_i = n \times \text{AM}, \qquad \frac{1}{R_{\text{parallel}}} = \sum \frac{1}{R_i}, \quad R_{\text{parallel}} = \frac{\text{HM}}{n}$$

So $3\,\Omega$ and $6\,\Omega$ give $9\,\Omega$ in series and $2\,\Omega$ in parallel, half their harmonic mean of $4\,\Omega$; equal resistors R give nR and R/n.



Series: voltages add.
Parallel: currents add.

KIRCHHOFF'S LAWS IN A TILING

Height is voltage, width is current

Kirchhoff's current law says the currents into any junction sum to the currents out, and his voltage law says the drops around any closed loop sum to zero. A rectangle of uniform conducting sheet, with current entering along its top and leaving along its bottom, has resistance $R = h/w$, so every square, of any size, is exactly $1\,\Omega$, with voltage equal to its height and current equal to its width. Squares side by side share their top and bottom edges and sit in parallel; squares stacked one on another sit in series. Each horizontal segment of a tiling becomes a junction at potential equal to its height, and each square becomes a $1\,\Omega$ resistor joining the segment on its top to the segment beneath it.

MOROŃ'S RECTANGLE, 1925

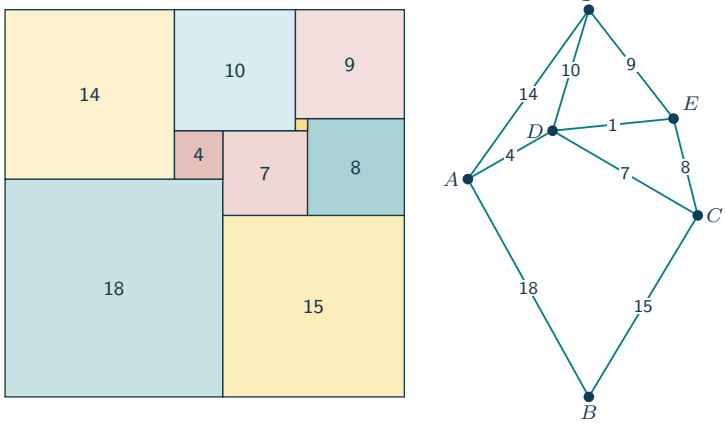
Nine squares, nine resistors

The 33×32 rectangle is tiled by squares of sides $18, 15, 14, 10, 9, 8, 7, 4, 1$, all different. Reading it as a network gives six junctions, the top T at height 32, the bottom B at 0, and four segments at heights $18, 15, 22, 23$, joined by nine $1\,\Omega$ resistors whose currents are the side lengths. Kirchhoff's current law holds at every junction because the widths above a segment fill the same length as the widths below it; at D, for instance,

$$10 + 1 = 4 + 7.$$

The whole network carries 33 units of current under a drop of 32, so its equivalent resistance is

$$R_{\text{eq}} = \frac{\text{height}}{\text{width}} = \frac{32}{33}\,\Omega$$



The tiling and its network, drawn with each junction at the height of its segment.

RUNNING IT BACKWARDS

Draw a network, solve it, get a tiling

Brooks, Smith, Stone and Tutte turned the dictionary round in 1940: choose a planar network of equal resistors, solve Kirchhoff's laws, and the currents are the sides of a squared rectangle. Feeding in a current equal to the number of spanning trees, 66 for Moroń's network, makes every current a whole number, here twice each side; distinct currents give a perfect rectangle, and the method went on to find perfect squared squares, down to Duijvestijn's 21 squares of 1978.

WHY THE CURRENTS ARE UNIQUE

Least energy picks the flow

Among all flows that respect the current law, the one Kirchhoff's voltage law selects is the one that dissipates the least power $\sum I^2 R$, Thomson's principle. With every $R = 1$ and the total current fixed at 33, the squares minimise the sum of their areas, $\sum s^2 = 33 \times 32 = 1056$, against every other split; a linear programme searches the corners of the same set of flows instead of its energy minimum.

Conventions. R is resistance in ohms, conductance is $1/R$, AM and HM are the arithmetic and harmonic means of the n resistances, and a squared rectangle is perfect when all its squares differ in size.